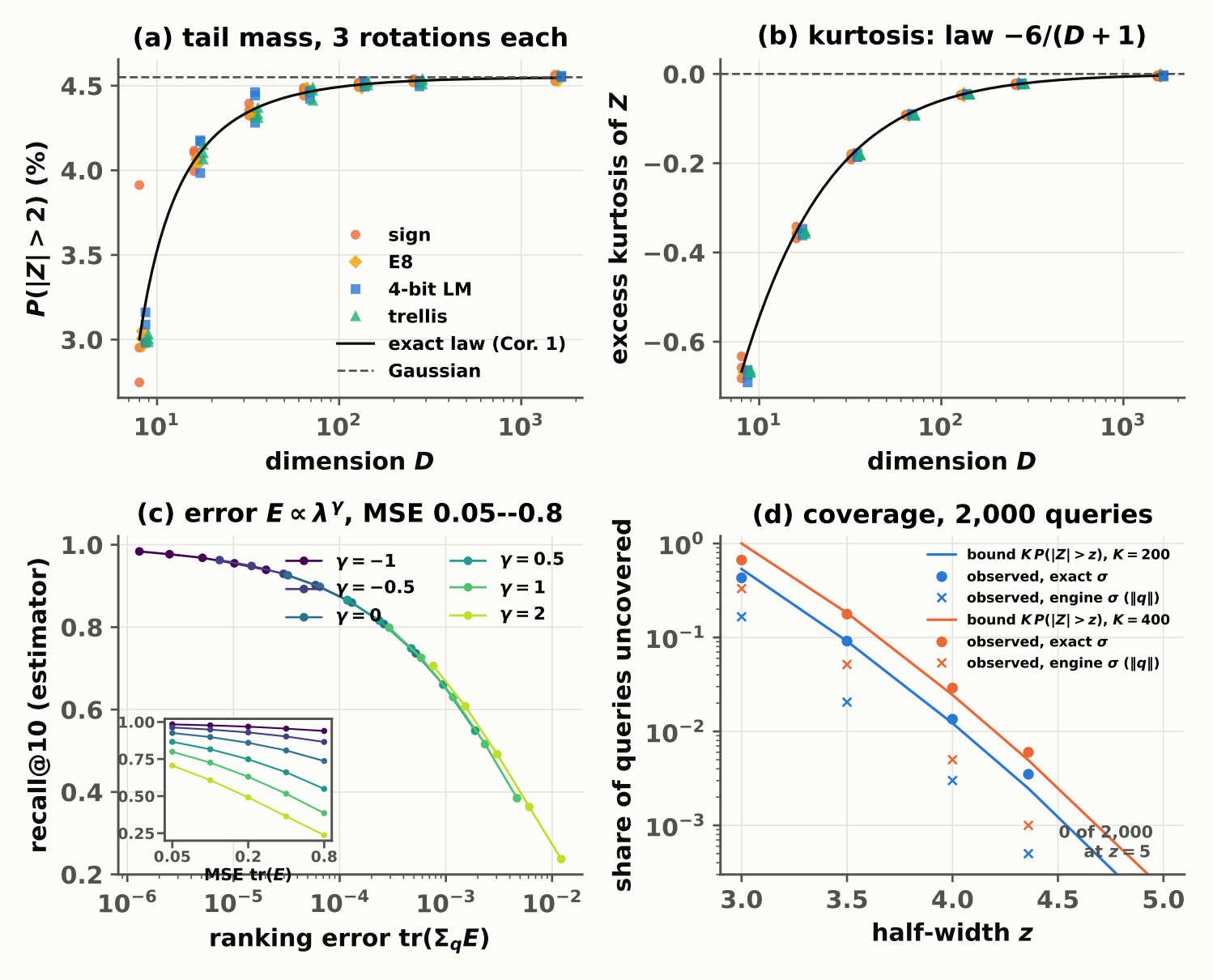
(a) tail mass, 3 rotations each
$P(|Z| > 2)$ (%)
4.5
4.0
3.5
3.0
sign
E8
4-bit LM
trellis
exact law (Cor. 1)
Gaussian
$10^1$
$10^2$
$10^3$
dimension $D$
(b) kurtosis: law $-6/(D+1)$
excess kurtosis of $Z$
0.0
−0.2
−0.4
−0.6
$10^1$
$10^2$
$10^3$
dimension $D$
(c) error $E \propto \lambda^\gamma$, MSE 0.05--0.8
recall@10 (estimator)
1.0
0.8
0.6
0.4
0.2
$\gamma = -1$
$\gamma = -0.5$
$\gamma = 0$
$\gamma = 0.5$
$\gamma = 1$
$\gamma = 2$
1.00
0.75
0.50
0.25
0.05
0.2
0.8
MSE tr($E$)
$10^{-6}$
$10^{-5}$
$10^{-4}$
$10^{-3}$
$10^{-2}$
ranking error tr($\Sigma_q E$)
(d) coverage, 2,000 queries
share of queries uncovered
$10^0$
$10^{-1}$
$10^{-2}$
$10^{-3}$
bound $KP(|Z| > z)$, $K = 200$
observed, exact $\sigma$
observed, engine $\sigma$ ($\|q\|$)
bound $KP(|Z| > z)$, $K = 400$
observed, exact $\sigma$
observed, engine $\sigma$ ($\|q\|$)
0 of 2,000
at $z = 5$
3.0
3.5
4.0
4.5
5.0
half-width $z$